

The Missing Minute

Phase, Memory and Waiting in “Every X Minutes” Claims

Caleb Gottfried

SNVRKOTICS LLC · snvrk@snvrkotics.com

Version 1.0 · October 2026

ABSTRACT

A claim that an event happens “every X minutes” seems to need two numbers, an interval and the minute it starts, yet only the first is ever given. We show that this paradox has an exact answer, and that it hides two quantities such claims always leave out. First, the start minute exists exactly when the process has a spectral line at frequency $1/X$, an atom in its Bartlett spectrum (Theorem 1). When it exists, the minute belongs to each realization: it is uniformly distributed across realizations, recoverable from event times with a closed-form standard error, and impossible to state in any summary of the process’s law. Poisson, renewal and pooled processes have no line, so they have no start minute. Second, we define *phase memory*, the number of events over which a schedule stays predictable. It is the interval noise spectrum seen through a Fejér filter, equals $1/(2\pi^2 F)$ for the long-run Fano factor F , and never fades exactly when the intervals are a schedule plus stationary errors (Theorem 3). Phase memory has a practical cost: applied to a process with memory but no line, the standard Rayleigh test for periodicity rejects at rate $\alpha^{1/\tau}$ instead of α , where $\tau \approx 2\ell$ when memory is long. We give a corrected test. Old Faithful’s 1985 record shows all of this at once. Its intervals alternate (lag-1 correlation -0.70 ; renewal rejected, $p < 10^{-4}$). It keeps time 2.6 times more precisely over two eruptions than over one (95% CI 2.2–3.0), because the two-interval filter cancels the alternation, much as a spin echo does. It remembers its phase for about five eruptions (95% CI 2.3–6.4), against the 1.4 a renewal model predicts. An apparent periodicity across the whole record (naive $p = 0.007$) vanishes once memory is accounted for ($p = 0.46$). A century of British coal-mining disasters shows the opposite type: no phase, no memory, and a mean wait three times the naive one. For any stationary process, the mean wait from a random moment is $(X/2)(1 + c^2)$, and a clock’s phase memory is bounded by the entropy it produces per tick: $\ell \leq \Delta s/4\pi^2$. Every “every X ” claim should come with two more numbers: its coefficient of variation and its phase memory.

Keywords: point processes; Bartlett spectrum; phase estimation; circular statistics; Rayleigh test; phase diffusion; renewal theory; Palm calculus; thermodynamic uncertainty relation; Old Faithful

1. Introduction

“A baby is born every few seconds.” “Someone is diagnosed every few minutes.” “Old Faithful erupts every hour and a half.” The form is everywhere because it turns an abstract total into something a reader can feel happening. The U.S. Census Bureau’s population clock, for example, expresses births, deaths and net migration each as one event every so many seconds [1]. The form also invites a question that is almost never asked. A train that leaves every ten minutes leaves at particular minutes, say :03, :13 and :23, so its

timetable needs an interval and an offset. A statistic that gives the interval alone is, read literally, half a timetable. We call the missing half the *phase*, and the impossibility of checking the claim without it the *phase problem*.

The phase problem looks like a quibble about wording. We argue that it opens onto two quantities that “every X ” claims systematically leave out, each of which changes what a reader should expect: whether the next event can be predicted at all, and how long they will wait for it. The paper makes six contributions.

- **An existence theorem.** The start minute exists exactly when the process has a spectral line at frequency $1/X$ (Theorem 1). For a stationary process it is then a random variable, fixed in each realization and uniform across them. This is the exact sense in which a statistic, which describes the law of the process, cannot say what minute. Pooling independent sources cannot create a line, and dilutes any line the sources have by the square root of their number.
- **Phase memory and a dichotomy.** The coherence $C(k)$ between events k apart is governed by the variance of the time spanned by k intervals. That variance is the interval noise spectrum seen through a Fejér filter (Proposition 3), and its growth sorts processes into scheduled, renewal and pooled types. Memory never fades exactly when that variance stays bounded, that is, when the process is a schedule plus stationary errors (Theorem 3). The memory length is a reparametrization of the classical long-run Fano factor, $\ell = 1/(2\pi^2 F)$.
- **A corrected test for periodicity.** Phase memory without a line makes event phases look clustered. The Rayleigh test then rejects at rate $\alpha^{1/\tau}$, where τ , the continuum spectrum at frequency $1/X$, is about 2ℓ when memory is long. We give an event-domain estimator of τ that restores the nominal size (Proposition 4).
- **A waiting law.** By Palm inversion, the mean wait from a random moment is $(X/2)(1 + c^2)$ for every stationary point process with finite second moments (Proposition 5). The result is classical in Palm calculus, but it is rarely used to check public claims.
- **A thermodynamic bound.** For a clock driven by a nonequilibrium Markov process, phase memory cannot exceed the entropy produced per tick divided by $4\pi^2$ (Proposition 6).
- **Two empirical studies.** In the 1985 Old Faithful record, a renewal model badly underestimates the geyser’s timekeeping, and the filter view explains why. The record’s apparent periodicity, however, is an artifact of memory. British coal-mining disasters, 1851–1962, show the pooled type, where the naive reading of “one every seven months” understates the typical wait threefold.

Section 2 reviews related work and Section 3 states the paradox. Sections 4 to 7 develop the theory, and Section 8 tests it by simulation. Sections 9 and 10 apply it to Old Faithful and to coal-mining disasters. Section 11 places phase memory alongside dephasing, oscillator noise and nonequilibrium thermodynamics, and Section 12 turns the results into guidance for writing such claims. Proofs are in Appendix A and simulation designs in Appendix B.

2. Related work

The ingredients are old; what is new is putting them together around a question of statistical communication. Spectral analysis of point processes begins with Bartlett [20] and is developed in Daley and Vere-Jones [14]. The serial-correlation structure of interval sequences, including the long-run Fano factor $F = c^2(1 + 2\Sigma r_j)$, is classical [21]. The neural-coding literature has shown that negatively correlated interspike intervals reduce long-run spike-count variability and improve coding [24]; our Old Faithful finding is the same phenomenon in a geyser. Phase diffusion is the standard description of noisy oscillators [26, 27]. Its discrete counterpart, the variance of accumulated timing error, is the quantity behind the Allan variance of clock metrology [25]. The dephasing of a qubit under Gaussian noise obeys the same exponential law as our Proposition 3, written with filter functions [28]; echo sequences suppress low-frequency noise by shaping those filters [29]. Periodicity searches in event data use the Rayleigh test and its relatives [8, 9, 11, 13]. In X-ray timing, power is routinely normalized to the local continuum, including the continuum created by detector dead time [33, 34]. Our corrected Rayleigh test is the event-domain form of that practice. The inspection paradox [3, 4] and its Palm-calculus form [15] give the waiting law. The thermodynamic uncertainty relation [30, 31] and its application to Brownian clocks [32] give the cost bound. We are not aware of earlier work that uses these tools to say when an “every X ” claim has a start minute, how long its schedule can be trusted, or why a famous geyser keeps better time in pairs.

3. The paradox

Let S be the statement “an event occurs every X minutes.” Its literal reading is a strictly periodic point process,

$$t_k = \varphi + kX, \quad k \in \mathbb{Z}, \quad (1)$$

with phase $\varphi \in [0, X)$. Checking S means checking event times against Eq. (1). That requires φ , and S withholds it:

A statement whose literal content is a timetable cannot be compared with any timetable, because half of the timetable is withheld. It sounds falsifiable, but as phrased it can't be checked.

Three readings of S are possible. Under the *periodic* reading, the phase is real but unstated. Under the *rate* reading, $X = T/N$ is a reciprocal average over a window, with no claim of regularity. Under the *aggregate* reading, events come from many independent sources (people, places, institutions) pooled into one count. We show that the paradox dissolves under all three, and that what remains is more interesting than the paradox.

4. When does the missing minute exist?

Let N be a stationary point process on $\mathbb{R}$ with intensity λ and finite second moments, and write $\omega = 2\pi/X$. Its Bartlett spectrum Γ is the measure for which $\text{Var} \int f dN = \int |\tilde{f}(u)|^2 \Gamma(du)$, where $\tilde{f}(u) = \int e^{iut} f(t) dt$ [14, 20].

Like any spectrum, Γ splits into lines (atoms) and a continuum. Define the empirical first Fourier coefficient at period X ,

$$\hat{c}_1(T) = (1/T) \int_0^T e^{i\omega t} N(dt) = (1/T) \sum_{t_k \leq T} e^{i\omega t_k}. \quad (2)$$

Theorem 1 (existence). As $T \rightarrow \infty$, $\hat{c}_1(T)$ converges in mean square to a random variable c_1 with $E|c_1|^2 = \Gamma(\{\omega\})$. Hence a start minute $\varphi = (X/2\pi) \arg c_1$ exists with positive probability if and only if the Bartlett spectrum has a line at ω ; otherwise $c_1 = 0$ almost surely. When the line exists, the law of φ is uniform on $[0, X)$.

The last sentence is the exact resolution of the paradox. Shifting time by s multiplies c_1 by $e^{-i\omega s}$. For a stationary process this leaves the law unchanged, so the law of the phase must be invariant under rotation, which means uniform. The start minute, when it exists, is a fact about the realization, not about the law. A statistic summarizes the law. So the statistic cannot say what minute, not because someone forgot to report it, but because the minute is not a property of what the statistic describes. It can, however, be read off the events themselves.

Corollary 1. (a) A Poisson process has $\Gamma(du) = (\lambda/2\pi)du$, so it has no line and no phase. (b) A stationary renewal process whose interval distribution is spread out has no line away from the origin [14], and hence no phase. (c) For independent processes, Bartlett spectra add, so pooling cannot create a line. If n independent schedules of equal rate each have resultant ρ at period X , their superposition has $ER^2 = \rho^2/n$: the line is diluted by $\sqrt{n}$ and vanishes in the Palm–Khinchine limit [5, 6]. (d) A schedule with independent errors, $t_k = \varphi + kX + \varepsilon_k$ with uniform φ , has a line of mass $\lambda^2 |Ee^{i\omega\varepsilon}|^2$ at ω .

Corollary 1 settles the aggregate reading. For pooled statistics such as “someone, somewhere, every X minutes,” the question “starting at which minute?” has no answer, because no realization singles out any minute. It also settles the rate reading, since $X = T/N$ contains no clock times at all. The periodic reading remains. If a phase exists, can an observer find it? Map each event time to an angle $\theta_k = \omega t_k \bmod 2\pi$ and form the mean resultant

$$z = (1/N) \sum_k e^{i\theta_k}, \quad R = |z|, \quad \hat{\varphi} = (X/2\pi) \arg z, \quad (3)$$

so that $R = X|\hat{c}_1|$ up to the ratio of N/T to $1/X$. This is the mean resultant of circular statistics [7, 13] and the basis of pulsar timing, where the arrival phase of the first recorded photon is likewise unknown [8, 9]. Let $\rho = E[\cos(\theta - \omega\varphi)]$ be the population concentration and $\rho_2 = E[\cos 2(\theta - \omega\varphi)]$.

Theorem 2 (recovery). For N independent event phases with $\rho > 0$, $\hat{\varphi}$ is consistent, with asymptotic standard error

$$SE(\hat{\varphi}) \approx (X/2\pi) \sqrt{[(1 - \rho_2) / (2N\rho^2)]}. \quad (4)$$

The Rayleigh statistic $Z = 2NR^2$ is approximately χ^2 with two degrees of freedom when $\rho = 0$, and noncentral χ^2 with noncentrality $2N\rho^2$ otherwise. Detecting the phase at $\alpha = 0.05$ with 80% power therefore requires about

$$N^* \approx 4.82 / \rho^2 \text{ events}. \quad (5)$$

Two cases cover most applications. A schedule with independent Gaussian timing errors of standard deviation σ has $\rho = \exp(-2\pi^2\sigma^2/X^2)$ and $\rho_2 = \rho^4$. A Poisson process whose rate is modulated as $\lambda_0[1 + \kappa \cos(\omega t - \omega\varphi)]$ has $\rho = \kappa/2$ and $\rho_2 = 0$. The paradox is therefore resolved in both directions. If the process has a line, the phase is not missing: it is an output of the data, not an input the claimant must supply. If it has no line, the phase does not exist. Theorem 2 assumes independent event phases. Section 6 shows what happens to the test, and how to repair it, when phases are dependent.

5. Phase memory

Theorem 1 asks whether a process has a phase over all time. A sharper practical question is local: if I know when one event happened, how well can I predict the phase of an event k events later? Define the *phase coherence* at lag k ,

$$C(k) = |E^0[e^{i\omega(t_{j+k} - t_j)}]|, \quad (6)$$

where E^0 is expectation under the Palm (event-stationary) distribution. In practice $C(k)$ is estimated by averaging over all pairs of events k apart. $C(k) = 1$ means the event k steps ahead falls at a perfectly predictable phase; $C(k) \approx 0$ means its phase is uniformly uncertain. Write $S_k = t_{j+k} - t_j$ for the time spanned by k consecutive intervals, so that $C(k) = |E^0 e^{i\omega S_k}|$ is the modulus of its characteristic function at ω . Let $f_I(v)$ on $[-\pi, \pi]$ be the spectral density of the interval sequence, so that its autocovariances are $\int e^{ijv} f_I(v) dv$.

Proposition 3 (phase diffusion). The span variance is the interval spectrum seen through a Fejér filter,

$$\text{Var}(S_k) = \int_{-\pi}^{\pi} f_I(v) F_k(v) dv, \quad F_k(v) = \sin^2(kv/2) / \sin^2(v/2), \quad (7)$$

and if S_k is approximately Gaussian,

$$C(k) \approx \exp(-\omega^2 \text{Var}(S_k) / 2). \quad (8)$$

Growth of $\text{Var}(S_k)$ therefore defines three types. **Type I (scheduled):** events are a fixed schedule plus independent errors; $\text{Var}(S_k) = 2\sigma^2$ for every $k \geq 1$, so $C(k) = \rho^2$ is constant and memory never fades.

Type II (renewal): intervals are independent with variance s^2 ; $\text{Var}(S_k) = ks^2$ and, exactly, $C(k) = |\psi(\omega)|^k$, where ψ is the interval characteristic function, so memory fades geometrically. **Type III (pooled):** in the Poisson limit $|\psi(\omega)| = (1 + 4\pi^2)^{-1/2} \approx 0.157$, so memory is gone after one event. In general $\text{Var}(S_k) \sim kD$, with long-run diffusion $D = 2\pi f_I(0) = s^2(1 + 2\sum_{j \geq 1} r_j)$, where the r_j are the interval autocorrelations. The *phase memory length*, the number of events over which coherence falls by a factor of e , is

$$\ell = 2 / (\omega^2 D) = 1 / (2\pi^2 F), \quad (9)$$

where $F = D/X^2$ is the long-run Fano factor of the counting process [21].

Equations (7) and (9) carry the paper's central intuition. Two processes can share the same X and the same interval variability s , yet have very different memory, because memory depends on whether consecutive intervals compensate one another ($r_1 < 0$, a long gap tends to follow a short one) or compound ($r_1 > 0$). A renewal model, which assumes every $r_j = 0$, can be wrong about predictability in either direction even when it fits the interval distribution perfectly. The filter makes the mechanism visible. Compensating intervals put their variance near $v = \pi$, the alternation frequency. For even k , $F_k(\pi) = 0$, so spans of an even number of intervals cancel the alternation; for odd k , $F_k(\pi) = 1$, so they keep it. A process with strongly alternating intervals should therefore keep better time over two events than over one, and its coherence should zigzag between even and odd lags. Section 9 finds exactly this in Old Faithful. The same arithmetic underlies the Hahn echo, in which a pulse halfway through a sequence of free evolutions cancels slow noise [29]; Old Faithful performs a natural echo every second eruption.

When does memory never fade? Proposition 3 suggests the answer, and Theorem 3 makes it exact.

Theorem 3 (memory dichotomy). Let the intervals be stationary under the Palm distribution, with mean X and finite variance. (i) $\sup_k \text{Var}(S_k) < \infty$ if and only if $t_j = t_0 + jX + \varepsilon_j - \varepsilon_0$ for some stationary sequence (ε_j) with finite variance; that is, if and only if the process is a schedule plus stationary errors. (ii) In that case, if (ε_j) is mixing, $C(k) \rightarrow |Ee^{i\omega\varepsilon_0}|^2$ as $k \rightarrow \infty$, and memory never fades. (iii) For a renewal process, $C(k) = |\psi(\omega)|^k \rightarrow 0$, unless every interval equals a fixed constant plus a multiple of X . (iv) More generally, if $\text{Var}(S_k) \rightarrow \infty$ and Eq. (8) holds, then $C(k) \rightarrow 0$.

Part (i) is a theorem about coboundaries of unitary operators [22, 23], here read as a statement about clocks. A process keeps time forever only if each event's error is measured against a fixed timetable rather than against the previous event. Every other process forgets, and ℓ says how fast.

6. Testing for a phase when memory is present

The Rayleigh test of Theorem 2 assumes independent phases. Phase memory violates that assumption without creating a line. Nearby events then sit at nearby phases, which makes R larger than chance, and the test mistakes memory for a phase.

Proposition 4 (Rayleigh test under phase memory). Let N be stationary with no line at ω , and let its Bartlett spectrum have a density γ continuous at ω . Then $E|\sum_{t_k \leq T} e^{i\omega t_k}|^2 / EN(0, T] \rightarrow \tau = 2\pi\gamma(\omega)/\lambda$.

In event time, when the series converges absolutely,

$$\tau = 1 + 2 \sum_{k \geq 1} \text{Re } E^0[e^{i\omega S_k}], \quad (10)$$

which equals $\text{Re}[(1 + \psi)/(1 - \psi)]$ for a renewal process, 1 for a Poisson process, and $\coth(1/2\ell) \approx 2\ell$ under Gaussian phase diffusion. If a central limit theorem holds for N , then $Z/\tau \rightarrow \chi^2_2$, so a naive Rayleigh test at level α rejects with probability tending to $\alpha^{1/\tau}$.

We estimate τ by truncating Eq. (10) at lag $L = N/10$, with the expectations replaced by averages over event pairs and X by T/N . The corrected p -value is $\exp(-Z/2\tau)$, which equals $p_{\text{naive}}^{1/\tau}$. Proposition 4 is the point-process counterpart of normalizing a periodogram by its local continuum, as X-ray astronomers do for red noise and dead time [33, 34]. What the phase-memory view adds is the event-domain estimator and the identity $\tau \approx 2\ell$: a test with N events has an effective sample size of about $N/2\ell$. A renewal process with $c = 0.2$ has $\tau \approx 2.7$, so a naive test on such a process, which has no phase at all, reports one at the 5% level about a third of the time (Section 8).

7. How long you wait

A reader told that something happens every 10 minutes, arriving at a random moment, expects to wait about 5. That expectation is right only for a perfect schedule. The general answer is the inspection paradox: a random moment is more likely to land in a long interval than in a short one [3, 4].

Proposition 5 (waiting law). For any stationary, ergodic, simple point process whose Palm interval distribution has mean X and coefficient of variation c , the mean wait from a random moment to the next event, and the mean length of the interval containing that moment, are

$$E[W] = (X/2)(1 + c^2), \quad E[I_{\text{straddle}}] = X(1 + c^2). \quad (11)$$

For renewal processes this is textbook material, and for stationary processes in general it follows from the Palm inversion formula [14, 15]. We state it because its generality is what makes it usable. It holds for scheduled processes with jitter, for processes with correlated intervals such as Old Faithful, and, by time averaging, for cyclostationary ones. We call $1 + c^2$ the *inspection multiplier*: the factor by which the true mean wait exceeds the naive $X/2$.

8. Simulations

All simulations use fixed seeds and are reproduced by the code accompanying this paper (Appendix B). Table 1 compares four processes that all average “one every 10 minutes” over 100,000 minutes.

Process	Events	T/N	c	R at X	p	Mean wait	Straddle	$(X/2)(1+c^2)$
A. Periodic, unknown phase	10,000	10.00	0.00	1.000	$< 10^{-300}$	5.01	10.00	5.00
B. Poisson	10,014	9.99	1.00	0.019	0.031	9.96	19.92	9.95
C. Pooled, 50 periodic sources	9,919	10.08	1.00	0.003	0.93	10.02	20.10	10.04
D. Poisson with daily cycle	9,926	10.07	1.12	0.009	0.44	11.36	22.77	11.37

Table 1. Four processes that all happen “every 10 minutes” (minutes throughout). The undisclosed phase of A was recovered to within 10^{-9} minutes.

Three results stand out. The rate is recovered in every case. A phase is found only where Theorem 1 says one exists. The nominally significant $p = 0.031$ for the Poisson process is the 1-in-20 false positive expected at $\alpha = 0.05$, and its concentration, $R = 0.019$, is negligible: a reminder that a single marginal test establishes nothing [12]. (For a Poisson process $\tau = 1$, so no correction applies.) And the waiting law predicts every row, including D, which is cyclostationary rather than stationary.

8.1 Phase recovery (Theorem 2)

Table 2 compares the standard error predicted by Eq. (4) with the spread of 400 replicate estimates for each design.

Design	N	ρ (theory)	R (mean)	SE, simulated	SE, Eq. (4)	$2N\rho^2$
Schedule, jitter $\sigma = 1$	100	0.821	0.821	0.101	0.101	135
Schedule, jitter $\sigma = 1$	1,000	0.821	0.821	0.030	0.032	1,348
Schedule, jitter $\sigma = 2$	100	0.454	0.460	0.235	0.243	41
Schedule, jitter $\sigma = 2$	1,000	0.454	0.456	0.075	0.077	412
Schedule, jitter $\sigma = 3$	100	0.169	0.185	0.769	0.665	6
Schedule, jitter $\sigma = 3$	1,000	0.169	0.172	0.200	0.210	57
Modulated Poisson, $\kappa = 1$	100	0.500	0.507	0.223	0.225	50
Modulated Poisson, $\kappa = 1$	1,000	0.500	0.502	0.071	0.071	500
Modulated Poisson, $\kappa = 0.5$	100	0.250	0.254	0.451	0.450	13
Modulated Poisson, $\kappa = 0.5$	1,000	0.250	0.251	0.128	0.142	125
Modulated Poisson, $\kappa = 0.2$	100	0.100	0.127	1.417	1.125	2
Modulated Poisson, $\kappa = 0.2$	1,000	0.100	0.102	0.388	0.356	20

Table 2. Phase standard errors (minutes, $X = 10$), 400 replicates per row.

The prediction is within about 10% of the simulated spread whenever $2N\rho^2 \gtrsim 10$, that is, once the phase is detectable at all. Below that it is optimistic, as an asymptotic formula should be. The mean resultant R is biased upward when $N\rho^2$ is small, by the familiar amount of order $1/\sqrt{N}$. For detection, Eq. (5) gives $N^* = 20, 54, 167$ and 482 events for $\rho = 0.5, 0.3, 0.17$ and 0.1 . In 2,000 replicates each, the Rayleigh test detected the phase with power 0.88, 0.83, 0.80 and 0.79, against the design target of 0.80.

8.2 The spectral line (Theorem 1)

Figure 1 shows Theorem 1 and Proposition 4 together. The scheduled process converges to its line. The Poisson process falls along $\sqrt{(\pi/4N)}$ (0.0049 at $N = 30,000$ against 0.0051 predicted). The renewal process sits higher by $\sqrt{\tau}$, with $\tau = 2.7$ as Eq. (10) predicts. A line-free model fitted to Old Faithful (Section 9) sits higher still, and at small N it is indistinguishable from the scheduled process. This is why a few hundred events from a process with memory can look periodic.

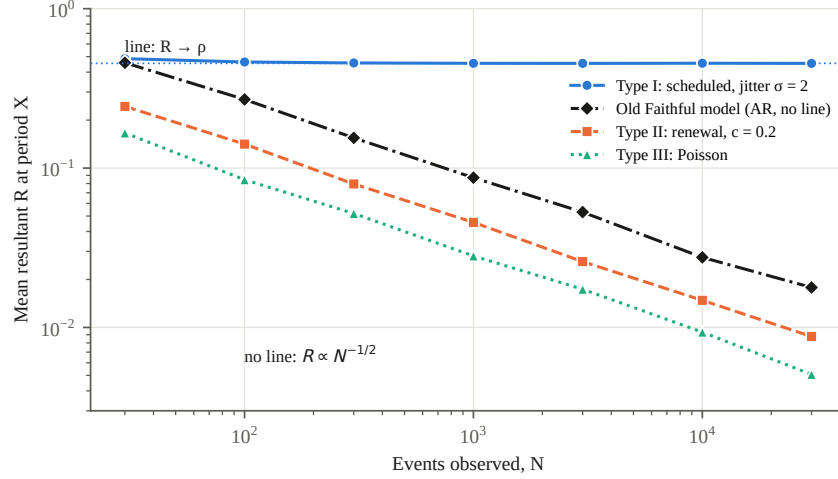


Figure 1. The line and the continuum. Mean resultant R at period $X = 10$ against the number of events observed (averages over 60–200 replicates per point). With a line (Type I), R converges to $\rho = 0.454$. Without one, R falls as $\sqrt{(\pi\tau/4N)}$ whatever the process. Processes with more memory sit higher (larger τ), but still fall to zero.

8.3 Phase memory (Proposition 3)

Figure 2 confirms the taxonomy. The scheduled process keeps $C(k)$ between 0.198 and 0.212 at every lag, against the predicted 0.206. The renewal process tracks $|\psi|^k$ (0.467, 0.220, 0.107 and 0.054, against 0.465, 0.216, 0.101 and 0.047) until it meets the noise floor. The Poisson process falls from 0.151 (theory 0.157) to the floor at once. The waiting law holds for the non-renewal Type I process as well. With $c = 0.284$, the simulated wait is 5.39 minutes against 5.40 predicted, and the straddling interval is 10.80 against 10.80.

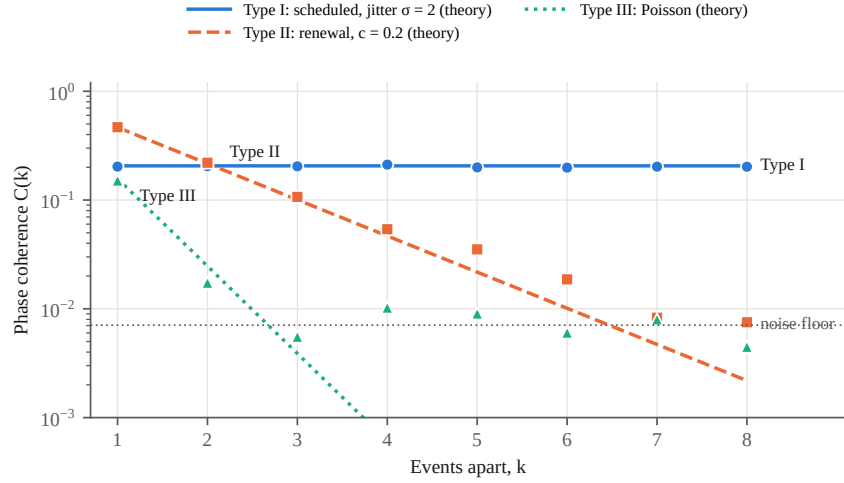


Figure 2. Phase coherence $C(k)$ for the three types, 20,000 events each ($X = 10$ minutes). Lines are theory and markers are simulation (circles, Type I; squares, Type II; triangles, Type III). Type I holds its coherence at $\rho^2 = 0.21$ at every lag. Type II (gamma intervals, $c = 0.2$) decays as $|\psi|^k$, with memory length 1.3 events. Type III reaches the noise floor ($1/\sqrt{N} = 0.007$) after one event.

8.4 The size of the Rayleigh test (Proposition 4)

Table 3 applies the naive and corrected Rayleigh tests to processes that have no line, only memory. Intervals follow a first-order autoregression around $X = 10$ with $c = 0.2$. Each run has $N = 300$ events, and X is estimated as T/N , as it would be in practice.

Interval correlation r_1	ℓ (theory)	mean $\hat{\tau}$	Naive size	Predicted $\alpha^{1/\tau}$	Corrected size
-0.7 (compensating)	7.2	10.1	0.75	0.74	0.024
-0.4	3.0	4.6	0.55	0.52	0.035
0 (renewal)	1.3	2.7	0.34	0.33	0.037
+0.4 (compounding)	0.5	2.2	0.27	0.26	0.028

Table 3. Rejection rates at nominal $\alpha = 0.05$ for processes with no phase, 2,000 replicates per row.

The naive test reports a nonexistent phase between a quarter and three quarters of the time. The closed form $\alpha^{1/\tau}$ predicts each rate to within 0.03. The corrected test holds its size, conservatively, at 2–4%. Even an ordinary renewal process, the textbook null for “random but regular,” is not a valid null for the naive test unless its τ is taken into account.

9. Old Faithful

Old Faithful in Yellowstone National Park is the canonical “every X minutes” phenomenon, famous precisely for its supposed regularity. We analyze the record of Azzalini and Bowman [16]: 299 consecutive intervals between eruptions, observed continuously from 1 to 15 August 1985, as distributed

with the MASS package for R [17, 18]. Event times are reconstructed by cumulating the intervals. Uncertainty intervals come from a sieve bootstrap [40]: 4,000 resamples from the fitted second-order autoregression of the intervals, which keeps their alternation. A moving-block bootstrap [39] gives similar intervals, but block joins break the alternation and bias it toward renewal, so we report it only as a check.

Quantity	Estimate	95% CI
Record	300 eruptions over 15.0 days	
Mean interval X (“every X minutes”)	72.3 min	
Interval SD; coefficient of variation c	13.9 min; 0.192	0.163–0.219
Lag-1 interval autocorrelation (lags 2–4)	−0.70 (+0.55, −0.45, +0.39)	−0.81 to −0.63
Mean wait from a random moment: naive $X/2$ · Eq. (11) · observed	36.2 · 37.5 · 37.4 min	
Interval containing a random moment: Eq. (11) · observed	75.0 · 74.9 min	
Precision gain over two eruptions, $c / CV(S_2)$	2.59	2.18–2.95
Coherence $C(1)$; $C(2)$	0.44; 0.65	0.29–0.50; 0.60–0.70
Long-run diffusion D per eruption (renewal: $s^2 = 192$)	51.7 min ²	34.9–70.4
Memory relative to renewal, s^2/D	3.7	1.5–4.8
Phase memory ℓ , Eq. (9) (renewal model: 1.4)	5.1 eruptions (6.2 h)	2.3–6.4 eruptions
Rayleigh test over all 15 days: naive · corrected ($\tau \approx 6.4$) · AR null	$p = 0.007 \cdot 0.46 \cdot 0.60$	

Table 4. Old Faithful, 1–15 August 1985. Sieve-bootstrap intervals; D by Bartlett-weighted autocovariances, lag 10 [42].

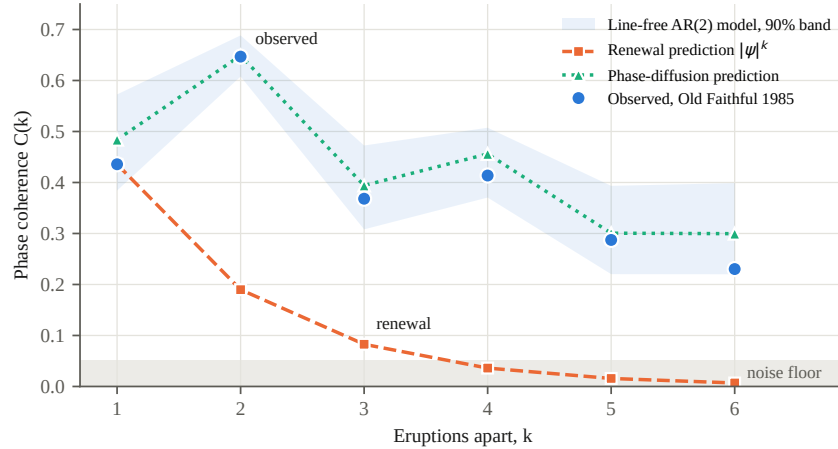


Figure 3. Old Faithful’s phase coherence by lag (circles). Shown against it are the renewal prediction $|\psi|^k$ (squares), the phase-diffusion law of Eq. (8) using the observed $\text{Var}(S_k)$ (triangles), and the 90% band of a line-free second-order autoregression fitted to the intervals (shaded). The grey band is the coherence expected from uniform phases with the same number of pairs (about 0.05).

Lag k (eruptions)	1	2	3	4	5	6
Observed $C(k)$	0.436	0.647	0.368	0.414	0.287	0.230
Renewal prediction $ \psi ^k$	0.436	0.190	0.083	0.036	0.016	0.007
Phase diffusion, Eq. (8)	0.484	0.649	0.394	0.456	0.301	0.300
Line-free AR(2) model, median	0.485	0.649	0.396	0.440	0.310	0.313
$\text{Var}(S_k), \min^2$ (renewal: $k \times 192$)	192	115	247	208	318	319

Table 5. Old Faithful phase coherence by lag. Uniform-phase noise floor ≈ 0.05 .

Five findings follow.

The waiting law holds on real data. With $c = 0.192$ the inspection multiplier is only 1.037, so a visitor waits 37.5 minutes on average rather than the naive 36.2; the observed 37.4 matches Eq. (11). Old Faithful earns its name in this sense: its multiplier is close to the scheduled ideal of 1.

Old Faithful is not a renewal process. Its intervals alternate, with autocorrelations of -0.70 , $+0.55$, -0.45 and $+0.39$ at lags 1 to 4: a short wait tends to be followed by a long one and vice versa. A surrogate test settles it [41]. Shuffling the same 299 intervals into random order 10,000 times, which keeps their distribution but destroys their order, never produced a two-eruption coherence above 0.32, against the observed 0.65 ($p < 10^{-4}$). Nor did it produce a ratio $\text{Var}(S_2)/\text{Var}(S_1)$ below 1.77, against the observed 0.60.

The mechanism is the eruption itself. The wait after an eruption tracks that eruption’s duration. Using the 221 eruptions whose durations were timed exactly (nocturnal durations were recorded only as short, medium or long and are excluded), the next wait is $34.9 + 10.6 \times \text{duration}$ minutes (correlation 0.90, residual SD 6.3 minutes). Durations themselves alternate (lag-1 correlation -0.66), so the waits alternate too. This is the recharge relation on which eruption predictions at the geyser have long been based [16, 44].

It keeps better time in pairs, and the filter says why. The time spanned by two eruptions varies less than the time spanned by one: $\text{Var}(S_2) = 115 \text{ min}^2$ against $\text{Var}(S_1) = 192$, a coefficient of variation of 0.074 against 0.192. That is 2.6 times as precise (95% CI 2.2–3.0). No renewal process can do this, because under renewal $C(2) = C(1)^2$ always. Proposition 3 explains the pattern. The intervals’ variance lies near the alternation frequency, which the even- k filters cancel and the odd- k filters keep. Hence the zigzag in Table 5, both in $\text{Var}(S_k)$ (192, 115, 247, 208, ...) and in coherence (0.44, 0.65, 0.37, 0.41, ...). For a visitor, “every 145 minutes, two eruptions at a time” is a better summary of Old Faithful than “every 72 minutes.”

Its phase memory is long, but it has no phase. Fitted to the same intervals, a renewal model predicts a memory length of 1.4 eruptions. The long-run diffusion is about a quarter of the renewal value, and Eq. (9) gives $\ell = 5.1$ eruptions (95% CI 2.3–6.4), about six hours. That is 3.7 times the renewal value (1.5–

4.8). The estimate is robust to the estimator of D : the slope of $\text{Var}(S_k)$ over lags 10–30 gives 5.4, and the autoregressive spectrum at zero gives 6.4. A line-free second-order autoregression, with nothing periodic in it, reproduces the observed coherence at every lag (Figure 3, Table 5). The full record does show a nominally significant clustering of phases ($R = 0.128$, naive $p = 0.007$). But that is exactly what Proposition 4 predicts memory alone will produce. With $\hat{\tau} = 6.4$ the corrected p is 0.46, and 60% of records simulated from the line-free model show at least as much clustering. Over fifteen days Old Faithful forgets its phase. What it has is not a start minute but an unusually long memory.

10. Coal-mining disasters

For contrast we take a pooled record: the dates of 191 explosions in British coal mines that killed ten or more people, from 15 March 1851 to 22 March 1962 [35, 37], as distributed with the R package boot [38]. On average there was one every 213 days, “a disaster every seven months.”

Quantity	Value
Mean interval X ; coefficient of variation c	213.4 days; 1.47
Mean wait from a random day: naive $X/2 \cdot \text{Eq. (11)} \cdot \text{observed}$	$107 \cdot 336 \cdot 337$ days
Interval containing a random day: $\text{Eq. (11)} \cdot \text{observed}$	$672 \cdot 672$ days
Phase coherence $C(1)$; $C(2)$ to $C(6)$ (noise floor 0.064)	0.21; 0.03–0.05
Rayleigh test at X : naive $\cdot$ corrected ($\hat{\tau} \approx 1.8$)	$p = 0.026 \cdot 0.14$
Before 1890: X, c (122 intervals) $\cdot$ from 1890: X, c (67 intervals)	116 days, 1.06 $\cdot$ 393 days, 1.14

Table 6. British coal-mining disasters with ten or more deaths, 1851–1962.

Here each of the paper’s three quantities takes its pooled value. There is no phase: the naive Rayleigh test reports one ($p = 0.026$), and the correction removes it. There is no memory: coherence is at the noise floor from the second event on. And the waiting law bites. With $c = 1.47$ the inspection multiplier is 3.15, so a random day waited on average 336 days for the next disaster, not the 107 that “one every seven months” suggests; the observed figure is 337. The large c has a known cause. The rate of disasters fell sharply around 1890 [35, 36], and within each era the intervals are close to exponential ($c \approx 1.1$). A single “every X ” for the whole century averages two regimes that never coexisted, which is Best’s point [2] in miniature: without its window, an “every X ” describes no period that actually happened.

11. A wider frame: dephasing, clocks and thermodynamics

Phase memory is not specific to geysers. Wherever events are counted against a clock, it reappears in some other vocabulary, and the correspondences carry results in both directions.

Dephasing and oscillator noise. Equations (7) and (8) are, term for term, the Gaussian dephasing law of a qubit, in which coherence decays as $\exp(-\frac{1}{2}\int S(\nu)F(\nu)d\nu)$ for noise spectrum S and a filter F set by the control sequence [28]. The Fejér filter is the filter of free evolution over k intervals. The long-run

diffusion D is the phase-diffusion constant of oscillator theory [26, 27], and $\text{Var}(S_k)/k^2$ is closely related to the Allan variance of clocks [25]. Pulsar timing uses the same structure: intrinsic spin noise is a phase diffusion, and timing models that treat it as white underestimate errors over long spans [45]. What the event-process view adds is the reading of D as a memory length in events, ℓ , and the Palm-calculus results that link it to waiting and to testing.

Proposition 6 (thermodynamic bound on phase memory). Suppose the events are the net completions of a cycle of a Markov jump process in a nonequilibrium steady state, producing on average Δs units of entropy (in units of k_B) per event. Then the long-run Fano factor satisfies $F \geq 2/\Delta s$, and hence

$$\ell \leq \Delta s / (4\pi^2). \quad (12)$$

Proposition 6 follows from the thermodynamic uncertainty relation [30, 31] and Eq. (9). A clock that is to stay in phase for ℓ ticks must dissipate at least $4\pi^2\ell k_B$ of entropy per tick. Table 7 checks the bound on Brownian ring clocks [32]: M -state cycles with forward rate f and backward rate b , for which $\ell = M(f - b)/[2\pi^2(f + b)]$ exactly. Memory grows with the number of states per tick, and the bound becomes tight near equilibrium. For Old Faithful the bound is vacuous: five eruptions of memory need only about 200 k_B per eruption, against a macroscopic dissipation larger by some twenty orders of magnitude. The bound binds only at the molecular scale, where a biochemical oscillator’s whole dissipation per cycle may be a modest number of ATP hydrolyses.

States per tick M	f/b	Entropy per tick Δs (k_B)	ℓ , exact	ℓ , simulated	Bound $\Delta s/4\pi^2$
1	20	3.0	0.046	0.046	0.076
5	20	15.0	0.23	0.23	0.38
20	20	59.9	0.92	0.89	1.52
20	3	22.0	0.51	0.48	0.56
100	3	109.9	2.53	2.67	2.78

Table 7. Phase memory of Brownian ring clocks against the bound of Eq. (12). Simulated ℓ from the long-run diffusion of 10^4 – 3.6×10^5 ticks.

Falsifiable predictions. The framework makes three predictions that could fail. (1) *Periodicity claims in event data.* Wherever the Rayleigh test, or a relative of it, is applied to event phases without a continuum correction (seismicity, chronobiology, phase locking of spikes), discoveries in data with measurable phase memory should be inflated at the rate $\alpha^{1/\tau}$. Reanalysis with $\tau\hat{s}$ should remove a predictable fraction of them. (2) *Compensating clocks.* Any event sequence with negatively correlated intervals, such as many neurons [24] and relaxation oscillators, should show $\text{Var}(S_2) < \text{Var}(S_1)$, $C(2) > C(1)$ and an even–odd zigzag in coherence, with magnitudes fixed by the interval spectrum through Eq. (7). (3) *Molecular clocks.* A

biochemical oscillator whose ticks are net cycle completions of a Markovian network cannot exceed Eq. (12). A measured phase memory above $\Delta s/4\pi^2$, with Δs computed from its free-energy consumption, would imply hidden non-Markovian structure or unaccounted dissipation.

The framework in six equations

Existence: $E|c_1|^2 = \Gamma(\{\omega\})$. A start minute exists if and only if the spectrum has a line at $1/X$, and it is then uniform across realizations.

Memory: $C(k) \approx \exp[-(\omega^2/2)\int f_I(v)F_k(v)dv]$: interval noise through a Fejér filter.

Memory length: $\ell = 2/(\omega^2 D) = 1/(2\pi^2 F)$, infinite exactly for a schedule with stationary errors.

Testing: $Z/\tau \sim \chi^2_2$ with $\tau = 2\pi\gamma(\omega)/\lambda \approx 2\ell$; naive size $\alpha^{1/\tau}$.

Waiting: $E[W] = (X/2)(1 + c^2)$ for any stationary process.

Cost: $\ell \leq \Delta s/4\pi^2$ for a Markovian clock.

12. Writing claims that can be checked

The analysis reduces to a short recommendation. An “every X minutes” claim states one number; the reader needs up to three.

1. Say “on average,” and give the count and window. “About N per year in [population], or one every X minutes on average” signals the rate reading, lets readers check the arithmetic, and reports what was actually measured. Best [2] documents how figures stripped of their denominator mutate as they are repeated, and the coal record (Section 10) shows how a window that spans two regimes produces an X that described neither.

2. Report the coefficient of variation c . It sets the expected wait through the inspection multiplier $1 + c^2$: 1 for a schedule, 2 for pooled events, more for a process whose rate changes.

3. When predictability matters, report the phase memory ℓ . It says how many events ahead the schedule can be trusted: forever for a timetable, about five eruptions for Old Faithful, one event or fewer for anything pooled. Anyone who tests for periodicity should report τ with the test.

Claim	X	c	Wait multiplier	Memory ℓ	Start minute?
Train timetable	10 min	≈ 0	1.0	∞	Yes; state it
Old Faithful, 1985	72 min	0.19	1.04	≈ 5 eruptions	No (corrected $p = 0.46$)
Coal-mining disasters, 1851–1962	213 days	1.47	3.15	< 1 event	No (corrected $p = 0.14$)
Pooled national statistic (theory)	seconds	≈ 1	≈ 2	< 1 event	No (Corollary 1)

Table 8. The three missing numbers for four “every X ” claims.

For genuinely scheduled processes, such as departures and automated jobs, the advice inverts: state the phase, because it exists and it is what the reader needs.

13. Limitations

Scope. The framework applies to processes whose events can be placed in time and whose intervals have finite second moments. It is not a claim that all systems are phase-diffusive. Poisson and pooled processes, including radioactive decay and the national statistics that motivate the paper, have neither phase nor memory, which is itself a result of Section 4.

Theory. Theorem 1 takes the first harmonic as the meaning of “phase at period X ”; a process with a line at 2ω but none at ω has structure at period $X/2$ that this definition does not call a phase. Theorem 2 assumes independent phases, and Eq. (4) is asymptotic, within about 10% once $2N\rho^2 \gtrsim 10$. Eq. (8) is exact only for Gaussian spans; at Old Faithful’s first lag, where the intervals are bimodal, it overstates coherence (0.48 against 0.44). Part (iv) of Theorem 3 assumes that approximation, and lattice-valued intervals are a genuine exception. Proposition 4 requires a central limit theorem, and the truncation $L = N/10$ in $\hat{\tau}$ is a heuristic, mildly conservative in our simulations (size 2–4%). Proposition 6 holds only for Markovian clocks whose ticks are a net current.

Data. The Old Faithful analysis rests on a single fifteen-day record from 1985. The geyser’s intervals respond to earthquakes and climate [43], so our findings describe that record, not the geyser today. The sieve bootstrap assumes a linear autoregression with independent residuals. With 299 intervals, ℓ is estimated only to within a factor of about 2. The coal record has one-day resolution and one zero interval, and its threshold of ten deaths is a reporting convention. Finally, we test no specific published statistic. The guidance of Section 12 is a proposal, not an evaluation of any existing figure.

14. Conclusion

“Every X minutes” seems to withhold half of a timetable. On inspection, the missing minute either exists in each realization and can be read from the events, or does not exist at all, and which of the two is decided by a single line in a spectrum. When it exists, no statistic can supply it, because statistics describe laws and the minute belongs to the realization. But the phrase also withholds two things that do exist and do matter. One is how irregular the process is, which sets how long anyone actually waits. The other is how long it remembers its own rhythm, which sets how far ahead it can be predicted, and how easily it can be mistaken for a clock. Old Faithful, the emblem of regularity, shows both sides. Judged by its memory, it is a far better clock than a renewal model allows, because each eruption corrects the last, and in pairs it keeps time like an echo. Judged by its phase, it has none to give. The missing minute was never the problem. The missing numbers were.

Code and data availability

Code, data and results: <https://github.com/snvrk/missing-minute>. Table 1: `code/sim.py`. Tables 2, 4 and 5, Figure 2 and the core Old Faithful analysis: `code/analysis.py`. Bootstrap intervals, surrogate and corrected tests, Tables 3, 6 and 7

and Figures 1 and 3: `code/robustness.py`. Figures: `code/figures.py`. All randomness is seeded, and every number is written to `results/results.json`, `results2.json` and `results3.json`. Data: `data/geyser.csv` (Azzalini–Bowman, via R’s MASS package) and `data/coal.csv` (Jarrett, via R’s boot package), both obtained from the Rdatasets collection [18].

Acknowledgments and disclosure

An AI writing tool assisted with drafting this paper and its analysis code; the author reviewed the result and is responsible for its content. No external funding was received, and the author declares no competing interests.

References

1. U.S. Census Bureau. *U.S. and World Population Clock*. <https://www.census.gov/popclock/>.
2. Best, J. *Damned Lies and Statistics: Untangling Numbers from the Media, Politicians, and Activists*. University of California Press, 2001.
3. Feller, W. *An Introduction to Probability Theory and Its Applications*, Vol. II, 2nd ed. Wiley, 1971.
4. Ross, S. M. *Introduction to Probability Models*, 11th ed. Academic Press, 2014.
5. Cox, D. R., and Isham, V. *Point Processes*. Chapman and Hall, 1980.
6. Khintchine, A. Y. *Mathematical Methods in the Theory of Queueing*. Griffin, 1960.
7. Mardia, K. V., and Jupp, P. E. *Directional Statistics*. Wiley, 2000.
8. Leahy, D. A., Darbro, W., Elsner, R. F., Weisskopf, M. C., Sutherland, P. G., Kahn, S., and Grindlay, J. E. “On searches for pulsed emission with application to four globular cluster X-ray sources.” *The Astrophysical Journal* 266 (1983): 160–170.
9. Buccheri, R., et al. “Search for pulsed γ -ray emission from radio pulsars in the COS-B data.” *Astronomy and Astrophysics* 128 (1983): 245–251.
10. Lewis, P. A. W., and Shedler, G. S. “Simulation of nonhomogeneous Poisson processes by thinning.” *Naval Research Logistics Quarterly* 26, no. 3 (1979): 403–413.
11. Lord Rayleigh. “On the resultant of a large number of vibrations of the same pitch and of arbitrary phase.” *Philosophical Magazine* 10 (1880): 73–78.
12. Wasserstein, R. L., and Lazar, N. A. “The ASA statement on p-values: context, process, and purpose.” *The American Statistician* 70, no. 2 (2016): 129–133.
13. Fisher, N. I. *Statistical Analysis of Circular Data*. Cambridge University Press, 1993.
14. Daley, D. J., and Vere-Jones, D. *An Introduction to the Theory of Point Processes*, Vol. I, 2nd ed. Springer, 2003.
15. Baccelli, F., and Brémaud, P. *Elements of Queueing Theory: Palm Martingale Calculus and Stochastic Recurrences*, 2nd ed. Springer, 2003.
16. Azzalini, A., and Bowman, A. W. “A look at some data on the Old Faithful geyser.” *Applied Statistics* 39, no. 3 (1990): 357–365.
17. Venables, W. N., and Ripley, B. D. *Modern Applied Statistics with S*, 4th ed. Springer, 2002.
18. Arel-Bundock, V. *Rdatasets: a collection of datasets originally distributed in various R packages*. <https://github.com/vincentarelbundock/Rdatasets>.
19. Gardner, W. A., Napolitano, A., and Paura, L. “Cyclostationarity: half a century of research.” *Signal Processing* 86, no. 4 (2006): 639–697.

20. Bartlett, M. S. "The spectral analysis of point processes." *Journal of the Royal Statistical Society B* 25, no. 2 (1963): 264–296.
21. Cox, D. R., and Lewis, P. A. W. *The Statistical Analysis of Series of Events*. Methuen, 1966.
22. Browder, F. E. "On the iteration of transformations in noncompact minimal dynamical systems." *Proceedings of the American Mathematical Society* 9 (1958): 773–780.
23. Lin, M., and Sine, R. "Ergodic theory and the functional equation $(I - T)x = y$." *Journal of Operator Theory* 10 (1983): 153–166.
24. Chacron, M. J., Longtin, A., and Maler, L. "Negative interspike interval correlations increase the neuronal capacity for encoding time-dependent stimuli." *Journal of Neuroscience* 21, no. 14 (2001): 5328–5343.
25. Allan, D. W. "Statistics of atomic frequency standards." *Proceedings of the IEEE* 54, no. 2 (1966): 221–230.
26. Demir, A., Mehrotra, A., and Roychowdhury, J. "Phase noise in oscillators: a unifying theory and numerical methods for characterization." *IEEE Transactions on Circuits and Systems I* 47, no. 5 (2000): 655–674.
27. Pikovsky, A., Rosenblum, M., and Kurths, J. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.
28. Cywiński, Ł., Lutchyn, R. M., Nave, C. P., and Das Sarma, S. "How to enhance dephasing time in superconducting qubits." *Physical Review B* 77 (2008): 174509.
29. Hahn, E. L. "Spin echoes." *Physical Review* 80 (1950): 580–594.
30. Barato, A. C., and Seifert, U. "Thermodynamic uncertainty relation for biomolecular processes." *Physical Review Letters* 114 (2015): 158101.
31. Gingrich, T. R., Horowitz, J. M., Perunov, N., and England, J. L. "Dissipation bounds all steady-state current fluctuations." *Physical Review Letters* 116 (2016): 120601.
32. Barato, A. C., and Seifert, U. "Cost and precision of Brownian clocks." *Physical Review X* 6 (2016): 041053.
33. van der Klis, M. "Fourier techniques in X-ray timing." In *Timing Neutron Stars*, NATO ASI Series C 262, 27–69. Kluwer, 1989.
34. Zhang, W., Jahoda, K., Swank, J. H., Morgan, E. H., and Giles, A. B. "Dead-time modifications to fast Fourier transform power spectra." *The Astrophysical Journal* 449 (1995): 930–935.
35. Jarrett, R. G. "A note on the intervals between coal-mining disasters." *Biometrika* 66, no. 1 (1979): 191–193.
36. Raftery, A. E., and Akman, V. E. "Bayesian analysis of a Poisson process with a change-point." *Biometrika* 73, no. 1 (1986): 85–89.
37. Hand, D. J., Daly, F., Lunn, A. D., McConway, K. J., and Ostrowski, E. *A Handbook of Small Data Sets*. Chapman and Hall, 1994.
38. Davison, A. C., and Hinkley, D. V. *Bootstrap Methods and Their Application*. Cambridge University Press, 1997.
39. Künsch, H. R. "The jackknife and the bootstrap for general stationary observations." *The Annals of Statistics* 17, no. 3 (1989): 1217–1241.
40. Bühlmann, P. "Sieve bootstrap for time series." *Bernoulli* 3, no. 2 (1997): 123–148.
41. Theiler, J., Eubank, S., Longtin, A., Galdrikian, B., and Farmer, J. D. "Testing for nonlinearity in time series: the method of surrogate data." *Physica D* 58 (1992): 77–94.
42. Newey, W. K., and West, K. D. "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix." *Econometrica* 55, no. 3 (1987): 703–708.
43. Hurwitz, S., Kumar, A., Taylor, R., and Heasler, H. "Climate-induced variations of geyser periodicity in Yellowstone National Park, USA." *Geology* 36, no. 6 (2008): 451–454.
44. Rinehart, J. S. *Geysers and Geothermal Energy*. Springer, 1980.

45. Shannon, R. M., and Cordes, J. M. “Assessing the role of spin noise in the precision timing of millisecond pulsars.” *The Astrophysical Journal* 725 (2010): 1607–1619.

Appendix A. Proofs and derivations

Theorem 1. Write $N(dt) - \lambda dt = \int e^{-iut} Z(du)$, the spectral representation of the centred process, where Z has orthogonal increments and $E|Z(du)|^2 = \Gamma(du)$ [14, Ch. 8]. Then $\hat{c}_1(T) - \lambda(e^{i\omega T} - 1)/(i\omega T) = \int K_T(\omega - u)Z(du)$, with $K_T(v) = (1/T)\int_0^T e^{ivt} dt$. The deterministic term tends to 0. Since $|K_T| \leq 1$, $|K_T(v)|^2 \leq 4/(Tv)^2$, $K_T(v) \rightarrow 1\{v = 0\}$, and Γ is translation-bounded, dominated convergence gives $E|(\int K_T(\omega - u) - 1\{u = \omega\})Z(du)|^2 \rightarrow 0$. Hence $\hat{c}_1(T) \rightarrow c_1 = Z(\{\omega\})$ in mean square, with $E|c_1|^2 = \Gamma(\{\omega\})$. If $\Gamma(\{\omega\}) = 0$, then $c_1 = 0$ almost surely. For uniformity, the shift θ_s maps $\hat{c}_1(T)$ to $e^{-i\omega s}\hat{c}_1(T)$ plus a term that vanishes as $T \rightarrow \infty$. Stationarity then makes the law of c_1 invariant under every rotation $e^{-i\omega s}$, and the only rotation-invariant law of $\arg c_1$ on $\{c_1 \neq 0\}$ is uniform. Corollary 1: (a) is the Poisson spectrum; (b) uses the renewal spectral density $(\lambda/2\pi)\text{Re}[(1 + \psi(u))/(1 - \psi(u))]$, which has no atoms away from 0 when the interval law is spread out [14]; (c) follows from additivity of covariance measures under independence, giving $E|c_1|^2 = \sum \lambda_i^2 \rho^2 = \lambda^2 \rho^2/n$ for equal rates $\lambda_i = \lambda/n$; (d) is the direct computation $c_1 = \lambda E[e^{i\omega \varepsilon}]e^{i\omega \varphi}$.

Theorem 2. Write $u_k = \theta_k - \omega \varphi$. The resultant’s component along the true direction has mean $N\rho$. The perpendicular component $\Sigma \sin u_k$ has mean 0 (by symmetry of the phase distributions considered) and variance $NE[\sin^2 u] = N(1 - \rho_2)/2$. To first order the angular error is the ratio of the perpendicular to the parallel component, with variance $(1 - \rho_2)/(2N\rho^2)$; multiplying by $X/2\pi$ gives Eq. (4) [7, 13]. Under $\rho = 0$ the two components are asymptotically independent normals with variance $N/2$, so $2NR^2 \rightarrow \chi^2_2$. Under $\rho > 0$ it is approximately noncentral χ^2_2 with noncentrality $2N\rho^2$. The noncentrality giving power 0.80 against the 0.95 quantile (5.99) is 9.63, so $N^* = 9.63/(2\rho^2) = 4.82/\rho^2$. For Gaussian jitter, $E[e^{iu}] = \exp(-\omega^2 \sigma^2/2)$, giving $\rho = \exp(-2\pi^2 \sigma^2/X^2)$ and $\rho_2 = \rho^4$. For the modulated Poisson process the phase density is $(1 + \kappa \cos u)/2\pi$, so $\rho = \kappa/2$ and $\rho_2 = 0$.

Proposition 3. $\text{Var}(S_k) = \Sigma_{a,b < k} \text{Cov}(I_a, I_b) = \int f_I(v) |\Sigma_{a < k} e^{iav}|^2 dv$, and $|\Sigma_{a < k} e^{iav}|^2 = \sin^2(kv/2)/\sin^2(v/2)$, which gives Eq. (7). Since F_k/k is the Fejér kernel times 2π , $\text{Var}(S_k)/k \rightarrow 2\pi f_I(0) = D$ when f_I is continuous at 0. For Gaussian S_k , $|Ee^{i\omega S_k}| = \exp(-\omega^2 \text{Var}(S_k)/2)$ exactly. Type I: $t_j = \varphi + jX + \varepsilon_j$ with independent errors gives $C(k) = |Ee^{i\omega \varepsilon}|^2 = \rho^2$ for every $k \geq 1$, with no Gaussian assumption. Type II: $Ee^{i\omega S_k} = \psi(\omega)^k$ exactly. For gamma intervals with shape a and mean X , $|\psi(\omega)| = (1 + (2\pi/a)^2)^{-a/2}$, with $a = 1$ the Poisson case. Eq. (9): $C(k) \approx \exp(-k\omega^2 D/2)$ falls by e over $\ell = 2/(\omega^2 D)$ events. By the central limit theorem for counting processes, $\text{Var } N(0, t] \sim tD/X^3$, so $F = D/X^2$ [21], and $\ell = 2X^2/(4\pi^2 D) = 1/(2\pi^2 F)$.

Theorem 3. (i) Let U be the unitary shift on the closed span of the centred intervals, and $g = I_0 - X$. Then $S_k - kX = \Sigma_{j < k} U^j g$, and by the theorem of Browder and of Lin and Sine [22, 23], $\sup_k \|\Sigma_{j < k} U^j g\| < \infty$ if and only if $g = (U - I)h$ for some h . Setting $\varepsilon_j = U^j h$, this says $I_j = X + \varepsilon_{j+1} - \varepsilon_j$, which after reindexing is the stated form. (ii) $C(k) = |Ee^{i\omega(\varepsilon_k - \varepsilon_0)}| \rightarrow |Ee^{i\omega \varepsilon_0}|^2$ by mixing. (iii) $|\psi(\omega)| = 1$ if and only if ωI is almost surely constant modulo 2π , that is, $I \in a + X\mathbb{Z}$ almost surely; otherwise $|\psi|^k \rightarrow 0$. (iv) is immediate from Eq. (8).

Proposition 4. With $f(t) = e^{i\omega t} 1_{[0, T]}(t)$, $E|fdN|^2 = \int |\tilde{f}(u)|^2 \Gamma(du) + |\lambda f|^2$. Here $|\tilde{f}(u)|^2 = \sin^2((u + \omega)T/2)/((u + \omega)/2)^2$, which integrates to $2\pi T$ and concentrates at $u = -\omega$, while $|\lambda f|^2$ is bounded. With no atom and γ continuous at $\pm\omega$ (and $\gamma(-\omega) = \gamma(\omega)$), the variance is $2\pi T\gamma(\omega)(1 + o(1))$, and dividing by $EN(0, T] = \lambda T$ gives τ . In event time, $E|\Sigma_{j \leq N} e^{i\theta_j}|^2 = N + 2\Sigma_{k < N} (N -$

$k)\text{Re } E^0 e^{i\omega S_k}$, and dividing by N gives Eq. (10) in the limit. For renewal, $E^0 e^{i\omega S_k} = \psi^k$, so $\tau = 1 + 2\text{Re}[\psi/(1 - \psi)] = \text{Re}[(1 + \psi)/(1 - \psi)]$. For Poisson, $\psi = 1/(1 - 2\pi i)$ and $\tau = 1$. Under Gaussian diffusion with $\text{Var}(S_k) = kD$ and $\omega X \in 2\pi\mathbb{Z}$, $E^0 e^{i\omega S_k} = q^k$ with $q = e^{-1/\ell}$, so $\tau = (1 + q)/(1 - q) = \coth(1/2\ell)$. The pseudo-variance $E[(fdN)^2]$ involves kernels concentrated at $u = -\omega$ and $u = +\omega$ simultaneously and is $o(T)$, so the resultant is asymptotically circular. Under a central limit theorem, $Z \approx \tau X^2$, and $P(Z > -2 \ln \alpha) \rightarrow \exp(\ln \alpha / \tau) = \alpha^{1/\tau}$.

Proposition 5. For a stationary ergodic simple point process with intensity $1/X$, the Palm inversion formula gives, for any non-negative functional f , $E[f(N)] = (1/X)E^0[\int_0^{I_0} f(\theta_s N)ds]$. Here I_0 is the interval following the event at the origin under the Palm distribution and θ_s is the shift [14, 15]. Taking f as the forward recurrence time gives $E[W] = (1/X)E^0[\int_0^{I_0}(I_0 - s)ds] = E^0[I_0^2]/(2X) = (X/2)(1 + c^2)$. Taking f as the length of the covering interval gives $E^0[I_0^2]/X = X(1 + c^2)$. No independence between intervals is used.

Proposition 6. For a Markov jump process in a nonequilibrium steady state with entropy production rate σ (units of k_B), every current J_t satisfies $\lim_{t \rightarrow \infty} [\text{Var}(J_t)/t]/[E(J_t)/t]^2 \geq 2/\sigma$ [30, 31]. The net number of cycle completions is a current, with $E(J_t)/t \rightarrow 1/X$ and $\text{Var}(J_t)/t \rightarrow F/X$, so $FX \geq 2/\sigma$, that is, $F \geq 2/(\sigma X) = 2/\Delta s$. Eq. (9) then gives $\ell = 1/(2\pi^2 F) \leq \Delta s/(4\pi^2)$. For the ring clock, the net position is a biased random walk with drift $f - b$ and variance rate $f + b$, so $F = (f + b)/[M(f - b)]$ and $\Delta s = M \ln(f/b)$. The bound reduces to $\ln r \geq 2(r - 1)/(r + 1)$ for $r = f/b \geq 1$, with equality as $r \rightarrow 1$.

Appendix B. Simulation design

Table 1: window 100,000 minutes. (A) $t_k = \varphi + 10k$, with $\varphi \sim U(0, 10)$ withheld from the estimator. (B) Homogeneous Poisson, rate $1/10$. (C) 50 periodic sources with periods $500 \times U(0.9, 1.1)$ and uniform phases, merged. (D) Thinning [10] of a Poisson process of rate $1.5/10$ with acceptance $(1 + 0.5 \sin(2\pi t/1440))/1.5$. Waiting measures use 200,000 uniform inspection moments. **Table 2:** 400 replicates per design, each with a fresh uniform phase; modulated designs draw event phases from $(1 + \kappa \cos u)/2\pi$ by rejection. Detection: 2,000 replicates per ρ at $N = N^*$. **Figure 2:** 20,000 events per type; Type I jitter $\sigma = 2$; Type II gamma intervals with shape 25 ($c = 0.2$); Type III exponential intervals. **Figure 1:** N from 30 to 30,000; 200 replicates per point up to $N = 3,000$ and 60 above. The Old Faithful model is the fitted AR(2) with resampled residuals, rescaled to $X = 10$. **Table 3:** Gaussian AR(1) intervals with mean 10 and SD 2, burn-in 200, $N = 300$ events, 2,000 replicates per design; τ truncated at $L = 30$. **Old Faithful:** AR order 2 by AIC (coefficients -0.627 and 0.107); 4,000 sieve-bootstrap and 4,000 block-bootstrap (block 20) resamples; 10,000 shuffle surrogates; 4,000 AR null records for the full-record test; 2,000 AR records for the coherence band. **Coal:** dates converted to days at 365.25 days per year; τ truncated at $L = 19$. **Table 7:** unicycle random walks with forward rate 1 and backward rate $1/r$, 4×10^5 to 2×10^6 steps; ticks are first passages to successive multiples of M ; ℓ from the Bartlett long-run variance (lag 20) of the tick intervals. Seeds: 20261003 (Tables 1–2, Figure 2) and 20261004 (everything else).